k explain the action of buffer solutions and carry out calculations on the pH of buffer solutions, e.g. making buffer solutions and comparing the effect of adding acid or alkali on the pH of the buffer

l use titration curves to show the buffer action and to determine K_a from the pH at the point where half the acid is neutralised

m explain the importance of buffer solutions in biological environments, e.g. buffers in cells and in blood (H_2CO_3/HCO_3^-) and in foods to prevent deterioration due to pH change (caused by bacterial or fungal activity).

4.8 Further organic chemistry

Related topics in Unit 5 will assume knowledge of this material.

Students will be assessed on their ability to:

1 Chirality

a recall the meaning of structural and E-Z isomerism (geometric/cis-trans isomerism)

b demonstrate an understanding of the existence of optical isomerism resulting from chiral centre(s) in a molecule with asymmetric carbon atom(s) and understand optical isomers as object and non-superimposable mirror images

c recall optical activity as the ability of a single optical isomer to rotate the plane of polarisation of plane-polarised monochromatic light in molecules containing a single chiral centre and understand the nature of a racemic mixture

d use data on optical activity of reactants and products as evidence for proposed mechanisms, as in S_N1 and S_N2 and addition to carbonyl compounds.

2 Carbonyl compounds

a give examples of molecules that contain the aldehyde or ketone functional group

b explain the physical properties of aldehydes and ketones relating this to the lack of hydrogen bonding between molecules and their solubility in water in terms of hydrogen bonding with the water

c describe and carry out, where appropriate, the reactions of carbonyl compounds. This will be limited to:

- i oxidation with Fehling's or Benedict's solution, Tollens' reagent and acidified dichromate(VI) ions
- ii reduction with lithium tetrahydridoaluminate (lithium aluminium hydride) in dry ether
- iii nucleophilic addition of HCN in the presence of KCN, using curly arrows, relevant lone pairs, dipoles and evidence of optical activity to show the mechanism
- iv the reaction with 2,4-dinitrophenylhydrazine and its use to detect the presence of a carbonyl group and to identify a carbonyl compound given data of the melting temperatures of derivatives
- v iodine in the presence of alkali.

3 Carboxylic acids

a give some examples of molecules that contain the carboxylic acid functional group

b explain the physical properties of carboxylic acids in relation to their boiling temperatures and solubility due to hydrogen bonding

c describe the preparation of carboxylic acids to include oxidation of alcohols and carbonyl compounds and the hydrolysis of nitriles

d describe and carry out, where appropriate, the reactions of carboxylic acids. This will be limited to:

   i reduction with lithium tetrahydridoaluminate (lithium aluminium hydride) in dry ether (ethoxyethane)

   ii neutralisation to produce salts, e.g. to determine the amount of citric acid in fruit

   iii phosphorus(V) chloride (phosphorus pentachloride)

   iv reactions with alcohols in the presence of an acid catalyst, e.g. the preparation of ethyl ethanoate as a solvent or as pineapple flavouring.

4 Carboxylic acid derivatives

a demonstrate an understanding that these include acyl chlorides and esters and recognise their respective functional groups, giving examples of molecules containing these functional groups

b describe and carry out, where appropriate, the reactions of acyl chlorides limited to their reaction with:

- i water
- ii alcohols
- iii concentrated ammonia
- iv amines

c describe and carry out, where appropriate, the reactions of esters. This will be limited to:

- i their hydrolysis with an acid
- ii their hydrolysis with a base, e.g. to form soaps
- iii their reaction with alcohols and acids to explain the process of trans-esterification and recall how it is applied to the manufacture of bio-diesel (as a potentially greener fuel) and low-fat spreads (replacing the hydrogenation of vegetable oils to produce margarine)

d demonstrate an understanding of the importance of the formation of polyesters and describe their formation by condensation polymerisation of ethane-1,2-diol and benzene 1,4-dicarboxylic acid.